

Safety portfolio and investigation pathways

Osterquill Toxicology Inc | Portland | 05/19/2025

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Start with the decision the package must support

- Early chemistry triage, dose selection and regulated safety conclusions require different investigation designs.
- Define the compliance scope before accepting test articles.
- Control validity, animal welfare oversight and traceability apply throughout; a non-GLP screen does not become GLP through report formatting.

Exploratory versus regulated work

- Non-GLP in vitro screens rank a test article cohort.
 - Exploratory rat range finding informs dose selection and tolerability.
 - Limit conclusions to the tested conditions and available exposure.
- GLP investigations follow an approved plan and controlled amendments.
 - Independent QA examines conduct and reporting.
 - The signed report identifies compliance scope and departures.

hERG (IKr) safety pharmacology

- Whole-cell patch clamp measures concentration-dependent current inhibition.
- Report potency in μM with evaluable concentration limits and cell-level qualifications.
- Reference-control acceptance precedes test article ranking.
- Interpret in relation to unbound exposure; screening alone cannot establish a clinical safety margin.

CYP safety interaction panel

- Evaluate inhibition across CYP1A2, CYP2C9, CYP2C19, CYP2D6 and CYP3A4.
- A fixed-concentration result identifies follow-up needs.
- CYP3A4 follow-up IC50 resolves concentration dependence for selected test articles.
- Record probe, reference control and historical comparison; do not infer mechanism from the initial panel.

HepG2 viability screen

- ATP-content response provides a cellular viability endpoint.
- Exposure duration, solvent and concentration coverage accompany each result.
- Check assay interference before interpreting reduced signal as toxicity.
- Use the screen to qualify later decisions, rather than extrapolating directly to whole-animal tolerability.

Genetic toxicology

- Bacterial mutation assay with and without S9 addresses activation-dependent responses.
- Mammalian-cell micronucleus assessment retains proliferation and cytotoxicity context.
- Vehicle and positive controls must support each treatment condition.
- Kenji Morita provides scientific interpretation; uncertain exposure-related findings require a defined repeat.

Exploratory rat range finding

- Rat DRF and recovery supports selection of doses and observation schedules.
- Express dose in mg/kg/day and retain dose-lot traceability.
- Individual-animal identifiers remain visible in observations, weight changes and pathology links.
- Exploratory findings guide design; they do not constitute a regulated GLP conclusion.

Animal welfare before and during exposure

- Define welfare observations, intervention criteria and escalation responsibilities before dosing.
- Promptly communicate a concerning observation to the responsible Study Director and designated welfare personnel.
- Document the observation time, intervention and subsequent condition on the animal card.
- Welfare action must not await the next scheduled QA gate.

Control validity: a scientific release condition

- Define acceptance criteria before acquiring the test article measurements.
- Retain rejected runs and explain repetition; present only accepted measurements as interpretable results.
- Flag concentration limitations and interference alongside the result.
- Do not suppress a qualification because a client needs a rapid decision.

QA gates follow the investigation

Scope agreement > Approved investigation plan > Conduct > Control validity and welfare review > Signed report > Archive transfer



Test article receipt > Dose-lot and identity reconciliation > Independent QA review

Roles and independence

- Dr N. Takamori: portfolio oversight and scientific escalation.
- Rachel Okafor: Toxicology Study Director; investigation conduct and conclusions.
- Lucia Moretti: independent GLP Quality Assurance and archive control.
- Kirsty Greenwood: sample management; Nathan Wade: pathology coordination.
- Scientific authors resolve findings; QA retains independent reporting of its observations.

Report and archive handover

- State the tested cohort, conditions, compliance scope and limitations early.
- Link summary findings to observation sheets, animal cards and dose lots.
- Retain original entries, amendments, control-validity evidence and report approval history.
- Separate exploratory recommendations from signed regulated conclusions.

Scope discussion with Osterquill

- Provide the intended decision, test article identity, available exposure information and material constraints.
- Agree assay endpoints and the requested compliance scope before scheduling.
- For in vivo work, establish dose rationale, welfare provisions and recovery objectives.
- Dr N. Takamori and the responsible Study Director will define the investigation and QA gates.